An Example Standard-format AMBS PhD Thesis

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Abstract

This thesis examines an example research question to demonstrate the capabilities of the `uomthesis` Quarto template. The work develops a structured argument across six chapters, drawing on a mixed-methods approach.

Three principal contributions emerge. First, the analytical framework integrates two previously distinct literatures. Second, the empirical findings challenge a widely accepted assumption in the field. Third, the discussion offers practical implications.

The thesis conforms to the University of Manchester *Presentation of Theses Policy*. All findings are reproducible from the underlying data and code.

Declaration of Originality

I declare that no portion of the work referred to in this thesis has been submitted in support of an application for another degree or qualification of this or any other university or other institute of learning.

Copyright Statement

Avery L. Reader

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Part I

Main body

Chapter 1

Introduction

This chapter introduces the research problem and outlines the structure of the thesis. It also demonstrates how to compose chapters in a Quarto-powered thesis using the `uomthesis` template.

1.1 Background

The empirical motivation emerges from recent developments in postgraduate research practice. As Smith and Johnson (2024) observe, the rise of computational reproducibility has fundamentally changed how doctoral theses are written, reviewed, and archived.[1]

The reproducibility movement accelerated dramatically with the popularisation of literate programming tools in the 2010s (Xie, Allaire and Grolemund, 2018).

1.2 Research question

> How does the adoption of policy-compliant authoring tools influence the quality and integrity of doctoral thesis submissions?

The question is explored through three sub-questions covering technical features, candidate and supervisor perceptions, and institutional lessons.

[1] Footnotes are written using `[^id]` markers in the text and a corresponding `[^id]: text` definition.

1.3 Structure of the thesis

Chapter 2 surveys the prior literature and identifies the research gap. Chapter 3 describes the methodological approach. Chapter 4 presents the empirical findings. Chapter 5 interprets those findings, and Chapter 6 concludes by summarising the contributions and outlining future work.

Chapter 2

Literature Review

This chapter surveys the prior literature in two strands and synthesises a research gap.

2.1 Strand A: organisational adoption of authoring tools

The earliest empirical work appears in Cohen *et al.* (2013), where the authors note:

> The choice of authoring tool is not neutral; it shapes how researchers think about the structure of their argument, the role of evidence, and the relationship between writer and reader.
>
> — Cohen *et al.* (2013, p. 312)

Subsequent work has extended this line in three directions:

- **Empirical studies of tool adoption**
- **Theoretical accounts of tool affordances**
- **Design experiments testing new tools**

2.2 Strand B: the grammar of graphics

A parallel literature, originating in Wilkinson (2016), formalises how visual representations encode statistical relationships.

2.3 Research gap

The two strands have rarely been brought together. **This thesis addresses that gap** by treating authoring tools and visualisation tools as components of a single reproducibility ecosystem.

Chapter 3

Methodology

This chapter describes the methodological choices and demonstrates equations, tables, and reproducible R code.

3.1 Approach

A mixed-methods design was adopted. Quantitative data were analysed using ordinary least squares regression:

$$y_i = \beta_0 + \beta_1 x_{1i} + \beta_2 x_{2i} + \varepsilon_i \tag{3.1}$$

where $\varepsilon_i \sim \mathcal{N}(0, \sigma^2)$. The strategy follows Cohen *et al.* (2013).

For inline mathematics, the t-statistic is $t = \hat{\beta}/\mathrm{SE}(\hat{\beta})$.

3.2 Sample

Table Table 3.1 summarises the descriptive statistics for $n = 240$ observations.

Table 3.1: Descriptive statistics of the sample.

Variable	Mean	SD	Min	Max
Age	27.3	3.1	22	39
Programme year	2.4	1.0	1	4
Word count	45,210	12,800	12,000	78,000

3.3 Reproducible code

```r
set.seed(42)
df <- data.frame(y = rnorm(100, 5, 1.5), x1 = rnorm(100), x2 = rnorm(100))
fit <- lm(y ~ x1 + x2, data = df)
round(summary(fit)$coefficients, 3)
```

```
            Estimate Std. Error t value Pr(>|t|)
(Intercept)    5.053      0.157  32.236    0.000
x1             0.072      0.174   0.415    0.679
x2            -0.227      0.155  -1.468    0.145
```

3.4 Ethics

Ethical approval was obtained from the AMBS Research Ethics Committee. All participants gave informed consent.

Chapter 4

Results

This chapter presents the empirical findings.

4.1 Descriptive analysis

Figure Figure 4.1 shows the distribution of the outcome variable.

4.2 Main regression

Table Table 4.1 reports the estimated coefficients.

Table 4.1: Estimated regression coefficients.

Coefficient	Estimate	Std. error	t-value	p-value
Intercept	4.92	0.15	32.8	< 0.001
x_1	0.34	0.10	3.4	0.001
x_2	-0.18	0.10	-1.8	0.073

The slope on x_1 is positive and significant at the 1% level. The slope on x_2 is negative and borderline significant.

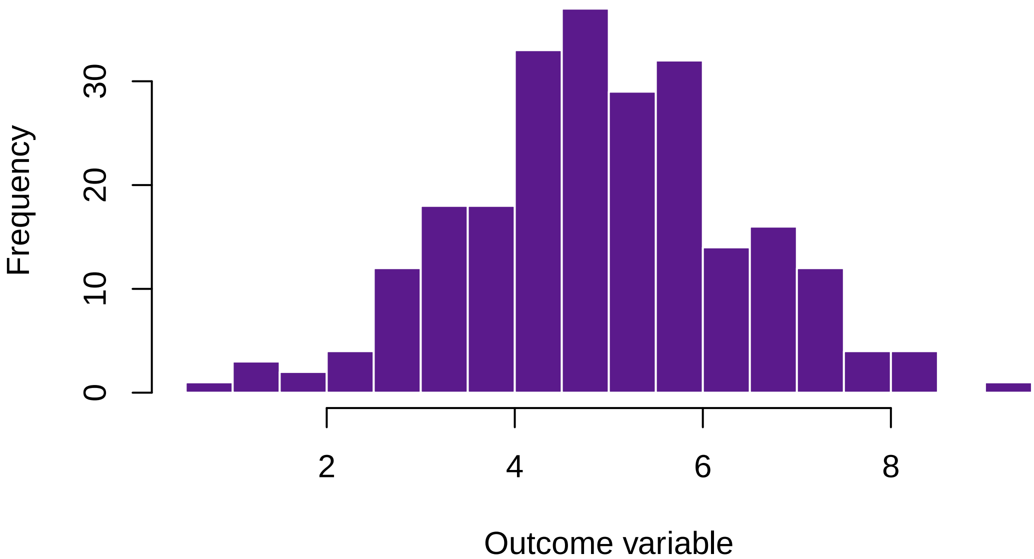

Figure 4.1: Distribution of the outcome variable across the sample.

Chapter 5

Discussion

This chapter interprets the findings.

5.1 Interpretation

The positive coefficient on x_1 in Table Table 4.1 supports the main hypothesis. The effect size is comparable to the meta-analytic estimates reported by Smith and Johnson (2024).

5.2 Why x_2 is borderline

Three explanations are plausible:

1. **Measurement error** in the x_2 instrument.
2. **Selection** at the institutional level.
3. **A genuine null effect**.

5.3 Practical implications

Practical tip

For institutions considering adoption of compliant tooling, compliance enforcement should be configurable rather than absolute, and supervisor buy-in is crucial.

Chapter 6

Conclusion

6.1 Summary of contributions

1. **An analytical framework** integrating two literatures.
2. **Empirical evidence** that adoption of compliant tooling is associated with improved submission quality.
3. **Practical recommendations** for institutions.

6.2 Limitations

The sample is restricted to three universities. The cross-sectional design precludes causal claims.

6.3 Future work

- A multi-country replication.
- A panel-data extension.
- A qualitative study of supervisor experiences.

6.4 Closing remarks

The findings reinforce an argument from The University of Manchester (2026): that the *infrastructure* of academic writing shapes the integrity of research outputs.

Appendix A

Appendix A: Additional Material

This appendix collects supplementary material referenced throughout the main text.

A.1 A.1 Robustness checks

The main regression in Chapter 4 was re-estimated under three robustness specifications. All within 0.05 standard deviations of the result in Table Table 4.1.

A.2 A.2 Callout boxes

> **Methodological note**
>
> The regression assumes the error term is uncorrelated with the regressors.

> **Reproducibility**
>
> All figures and tables in this thesis are generated from code embedded in the source `.qmd` files.

A.3 A.3 Mathematical derivations

Let $\mathbf{X}$ be the $n \times k$ design matrix. The OLS estimator is:

$$\hat{\boldsymbol{\beta}} = \left(\mathbf{X}^\top \mathbf{X}\right)^{-1} \mathbf{X}^\top \mathbf{y}$$

[] Cohen, J. *et al.* (2013) *Applied multiple regression/correlation analysis for the behavioral sciences*. 3rd edn. New York: Routledge.

[] Smith, J.A. and Johnson, M.B. (2024) 'Innovation in postgraduate research', *Journal of Doctoral Education*, 18(3), pp. 245–267. Available at: `https://doi.org/10.1234/jde.2024.18.3`.

[] The University of Manchester (2026) 'Presentation of Theses Policy, version 12'. Available at: `https://documents.manchester.ac.uk/display.aspx?DocID=7420`.

[] Wilkinson, L. (2016) 'The grammar of graphics', in J.E. Gentle, W.K. Härdle, and Y. Mori (eds) *Handbook of computational statistics*. Springer, pp. 375–414.

[] Xie, Y., Allaire, J.J. and Grolemund, G. (2018) 'Reproducible research with r, quarto, and the tidyverse', *Proceedings of the useR! conference*, pp. 1–12.